

Section-A (25 Marks)

1. What are the steps of a RCC singly reinforced beam design using limit state method? [5]

Answer:

Singly reinforced concrete section can be defined as the concrete section in which the main reinforcement bars are provided only in the tension zone (bottom part of beam section) as the concrete is weak in tension and prone to cracks and failures.

Steps of RCC singly reinforced beam design using limit state method:

i. Calculate factored load:

Dead load and live load to be subjected to the beam section is converted into factored load.

$$w_u = 1.5(w_D + w_L)$$

where,

$$w_D = \text{Dead Load } w_L = \text{Live Load}$$

ii. Determine factored Bending moment:

For a simply supported beam,

$$M_u = \frac{w_u L^2}{8}$$

For a cantilever beam,

$$M_u = \frac{w_u L^2}{2}$$

iii. Assume beam dimensions:

$$\text{Assume } \frac{l}{d} = 12$$

Then, we determine effective depth of beam section.

$$d = D - \text{cover} - \frac{\phi}{2}$$

$D = \text{overall depth,}$

$\phi = \text{diameter of tension bar}$

iv. Check limiting moment of resistance:

For Fe 415 steel,

$$M_{u, \text{lim}} = 0.138 f_{ck} b d^2$$

General expression:

$$M_{u, \text{lim}} = 0.36 f_{ck} b x_{u, \text{max}} (d - 0.42 x_{u, \text{max}})$$

If $M_u \leq M_{u, \text{lim}}$, then the section is singly reinforced.

v. Calculate required effective depth:

$$d = \sqrt{\frac{M_u}{0.138 f_{ck} b}}$$

vi. Determine natural axis depth:

Using equilibrium,

$$0.36 f_{ck} b x_u = 0.87 f_y A_{st}$$

Therefore,

$$x_u = \frac{0.87 f_y A_{st}}{0.36 f_{ck} b x_u}$$

vii. Calculate area of tension of steel:

$$M_u = 0.87 f_y A_{st} (d - 0.42 x_u)$$

Substituting x_u and solving gives A_{st} .
Alternatively,

$$A_{st} = \frac{M_u}{0.87 f_y (d - 0.42 x_u)}$$

viii. Provide reinforcement:

Number of bars:

$$n = \frac{A_{st}}{\frac{\pi}{4} \phi^2}$$

provide suitable bar arrangement.

ix. Check minimum and maximum reinforcement:

$$A_{st, min} = \frac{0.85 b d}{f_y}$$

Ensure $A_{st} \geq A_{st, min}$

$$A_{st, max} = 0.04 b D$$

Ensure $A_{st} \leq A_{st, max}$

x. Check shear strength:

Nominal shear stress:

$$\tau_v = \frac{V_u}{b d}$$

where,

$$V_u = \frac{w_u L}{2}$$

Compare τ_v with design shear strength τ_c from IS 456 tables.

If,

$$\tau_v > \tau_c$$

Then, the shear carried by stirrups:

$$V_{us} = V_u - \tau_c b d$$

Spacing of vertical stirrups:

$$s_v = \frac{0.87 f_y A_{sv} d}{V_{us}}$$

xi. Check deflection:

$\frac{L}{d} \leq$ permissible $\left(\frac{L}{d}\right)$ using modification factors from code.

xii. Check development length:

$$L_d = \frac{\phi \sigma_s}{\tau b d}$$

Ensure that the available anchorage length is greater than L_d .

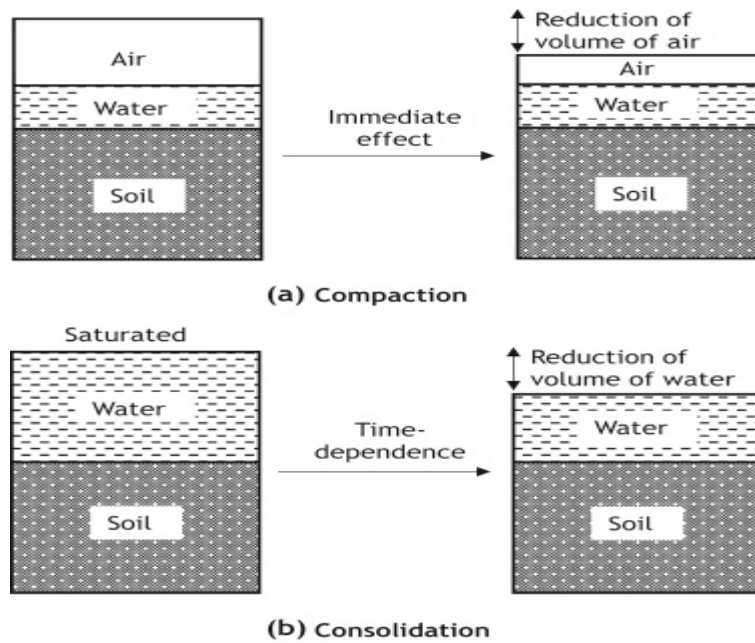
2. What is soil compaction? How is it different from consolidation? What are the factors that affect compaction? [1+2+2=5]

Answer:

Soil compaction is the process of densification of soil by removing the air voids present in the soil particles under the action of mechanical energy. Compaction is measured in terms of dry density.

Principle of soil compaction:

Required quantity of water is sprinkled in the soil before compaction such that water wets the surface of soil and act a lubricant between soil particles, reduce the friction so that most of the applied mechanical energy is used to make the soil particles closer.



Similarly, consolidation is the process of compression of water voids due to which expulsion of water from the void occurs under the effect of applied load.

The differences between soil compaction and consolidation are given in points.

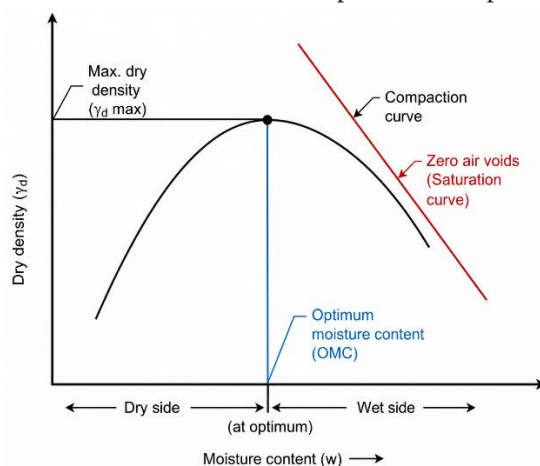
	Compaction		Consolidation
1.	Densification of soil under the action of mechanical energy by reduction of air voids.	1.	Compression of water voids under the effect of sustained applied load by reduction of water voids.
2.	Compaction is immediate process.	2.	Consolidation is slow and time-dependent process.
3.	Compaction increases dry density and strength of soil.	3.	Consolidation causes settlement and increase in effective stress.

Factors affecting compaction:

There are several factors affecting compaction. They are described in brief:

i. Moisture content of soil:

Maximum dry density of the soil is achieved at Optimum Moisture Content (OMC). Decrease in moisture content than OMC, makes it dry of optimum compaction while increase in moisture content than OMC makes it wet of optimum compaction.



ii. Amount of energy applied:

Larger maximum dry density can be achieved even at lower OMC, by application of larger magnitude of energy.

iii. **Types of soil:**

Coarser type of soil achieves higher dry density than finer soil type.

3. **What do you mean by structural irregularity in designing earthquake resistant building? Briefly illustrate the different aspects of vertical irregularity and mass irregularity. What is the importance of ductile detailing for the earthquake resistant building construction? [1+4+5=10]**

Answer:

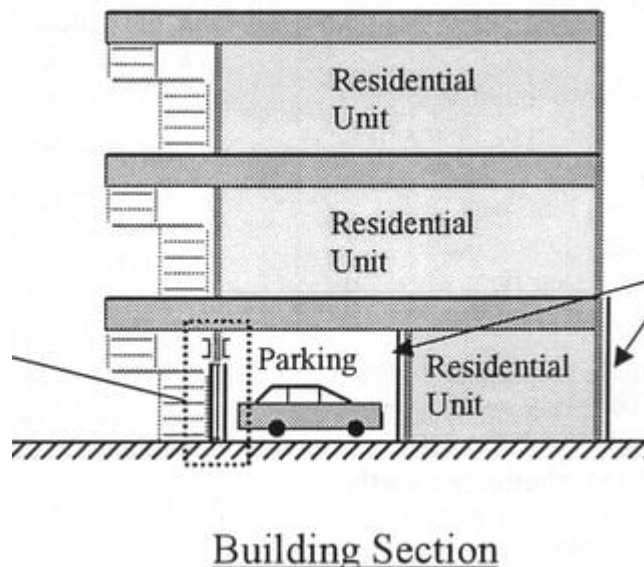
Structural irregularity can be defined as a condition in which the distribution of a building's mass, stiffness, strength, or geometry is not uniform along its height or plan. Due to this non-uniformity, the building does not respond evenly to earthquake forces, leading to stress concentration and increased vulnerability to damage.

Vertical irregularity is the structural irregularity that occurs when there is a sudden change in stiffness, strength, mass or geometry of a building along its height causing uneven distribution of seismic forces and lead to severe damage during earthquakes.

Aspects of vertical irregularity:

i. **Stiffness irregularity -soft storey:**

- lateral stiffness is less than 70% of the storey above or less than 80% of average lateral stiffness of 3 storey's above.
- Commonly occurs when the ground floor is left open for parking or commercial purposes.



ii. **Strength irregularity- weak storey:**

- lateral strength is less than 80% of a storey above.
- Caused by fewer columns, smaller members or inadequate reinforcements.

iii. **Structural irregularity- extreme soft storey:**

- Lateral stiffness is less than 60% of the storey above or less than 70% of the average stiffness in three storey's above.

iv. **In-plane discontinuity:**

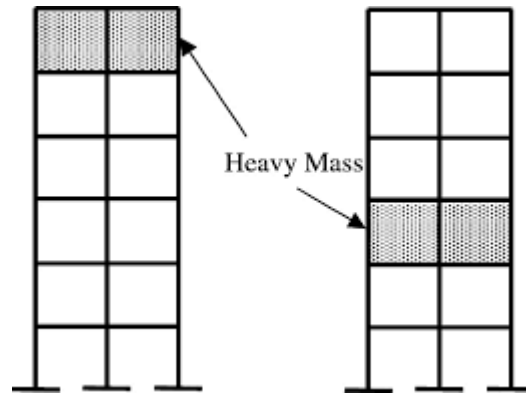
- It causes abrupt discontinuity in floors, diaphragms, shear walls or bracing systems.

Mass irregularity is the structural irregularity that exists when the seismic weight of any storey is greater than 200% of that of adjacent storey. It is caused by heavy floors, mechanical equipment floor, water tank at roof levels and so on.

Effects of mass irregularity:

- a. Increases seismic inertia forces at the heavy floor.

- b. Increases torsional response and vibration.
- c. Large storey shear and overturning moments.



Hence, vertical irregularities refer to abrupt change in stiffness, strength, geometry or load-resisting elements along the height of building while mass irregularities refer to sudden increase in floor mass at one level. Both irregularities require special analysis and detailing in earthquake design.

Ductile detailing is the provision of special reinforcement details in structural members such as beams, columns, joints, and shear walls so that they can undergo large deformations during an earthquake without sudden failure.

Importance of ductile detailing:

i. Improves safety of structure:

Ductile detailing helps the structure to withstand earthquake forces without collapsing.

It improves the structural integrity of structure even when large dynamic forces are subjected.

ii. Prevents sudden brittle failure:

Concrete is brittle in nature and may fail under large seismic and dynamic loads. However, ductile detailing increases the ductility and flexibility of structures and allows the members to deform significantly before failure.

iii. Enhances energy dissipation during earthquake:

Properly detailed reinforcement helps the structure for uniform dissipation of energy through controlled inelastic deformation when earthquake load is subjected to the structures.

iv. Enhances performance of beam-column joints:

Proper detailing also prevents cracking and crushing through the joint as it remains intact and transfer forces effectively. Hence, detailing enhances the performance of beam-column joints.

v. Reduces loss of life and property:

Ductile detailing helps prevent the collapse of structure even though it suffers severe damage. This property of detailing helps to buy the time to vacate the buildings or structure before the failure of structure, which prevents loss of life and property.

4. What is the purpose of site investigation? What are the different in-situ test used in site investigation? How bearing capacity is calculated using N-value for isolated footing, mat, pile and well? [2+4+4=10]

Answer:

Site investigation is the process of acquiring geological, geotechnical and other relevant information which might affect on the design, construction and environment of any civil engineering structure. It is also called site exploration.

Purposes of site investigation:

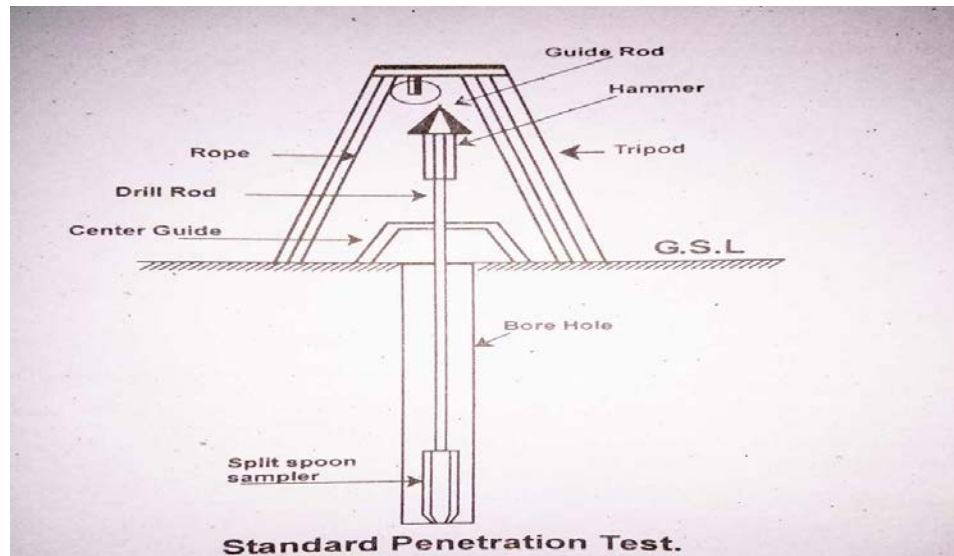
- i. Site investigation helps to determine vertical and lateral extension of each soil layer in the site.

- ii. It helps to determine engineering properties of each soil layers through series of in-situ test and lab tests.
- iii. It also helps to determine the position of groundwater table.
- iv. Site investigation also helps to determine index properties of each soil layers through series of lab tests.

In-situ tests are those tests used for evaluation of soil at their natural location without extracting samples for laboratory analysis. There are different in-situ tests used in site investigation. They are described below:

i. Standard penetration test:

- SPT tests is one of the most commonly used field tests.
- The test is conducted in a borehole using a split-spoon sampler from where N-value is determined.
- The number of blows required to drive the sampler 300 mm is called SPT N-value.

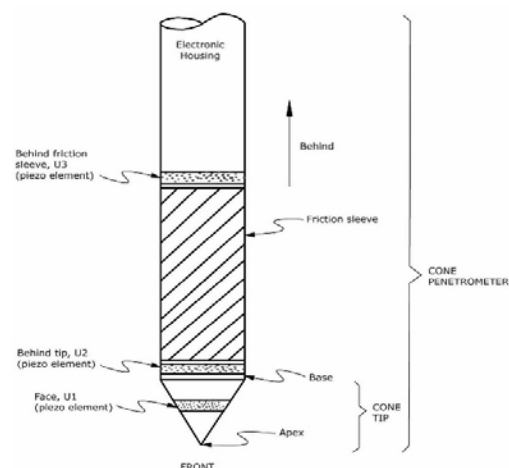


ii. Cone penetration test:

In cone penetration test, a cone is pushed into ground at a constant rate of 2 cm per second.

It measures cone resistance and sleeve friction.

The number of blows of hammer required to penetrate a cone to 300 mm depth is called CPT N_c value.

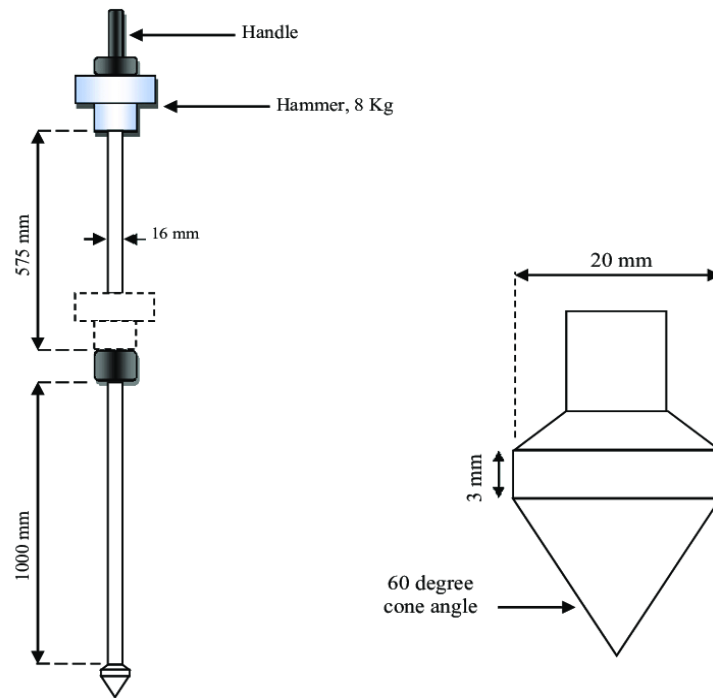


iii. Dynamic cone penetration test:

DCPT is similar to SPT but is carried out without a borehole.

It is useful for rapid assessment of subsoil conditions in boulder, gravel and rock layer.

The method is suitable in hilly area.



Section-B (25 Marks)

5. Explain in brief Bernoulli's equation and its application. [3+2=5]

Answer:

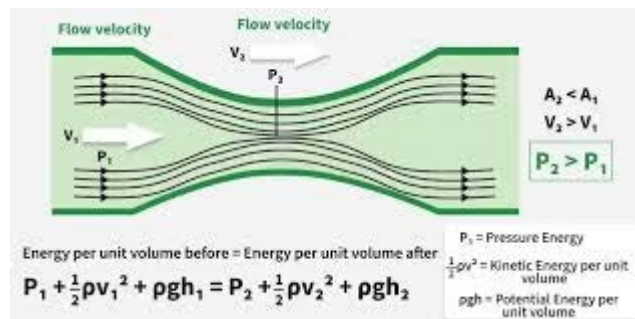
Bernoulli's principle is based on the principle of conservation of energy applied to the fluid. It states that for an ideal fluid flowing steadily through a pipe, the total energy remains constant along a streamline. The sum of pressure energy, kinetic energy, and potential energy per unit weight of fluid remains constant.

Bernoulli's Equation

$$\frac{P}{\gamma} + \frac{V^2}{2g} + z = \text{Constant}$$

or between two points,

$$\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + z_2$$



Where:

P = Pressure of fluid (N/m²)

γ = Specific weight of fluid (N/m³)

V = Velocity of fluid (m/s)

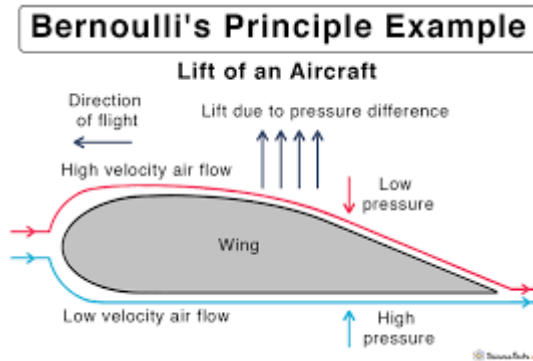
g = Acceleration due to gravity (9.81 m/s²)

z = Elevation head above a datum (m)

The applications of Bernoulli's principle:

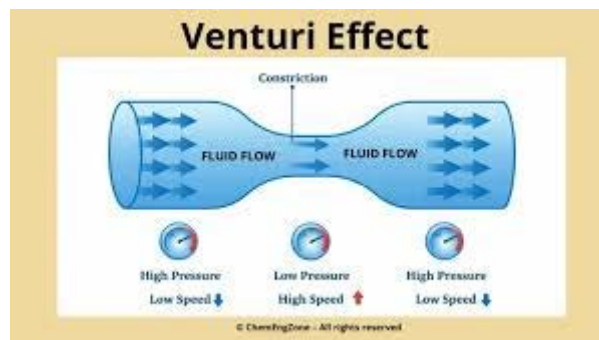
i. Aircraft wings:

Curved surface of aircraft wings forces air to move faster over top surface creating low pressure than the lower flat surface. This pressure difference creates the lift in the wings.



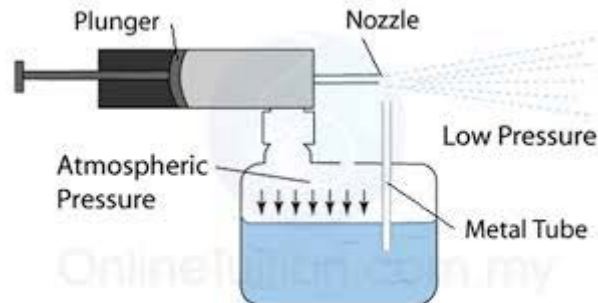
ii. Flow meters:

Devices such as orificemeter, venturimeter uses Bernoulli's equation to determine fluid flow rates in pipe.



iii. Nozzle:

Air blown across the top of a tube moves at high speed so that pressure above the tube decreases. The liquid is then pushed up the tube and sprayed as fine droplets.



6. Define Duty, Delta and Base period of a crop. Derive the relation between them. [4+6=10]

Answer:

Duty:

Duty of water is defined as the number of hectares that can be irrigated by constant supply of water at the rate of 1 cumecs throughout the base period. The duty of water is not constant but it varies with various factors like soil conditions method of flowing method of application of water.

Delta:

If the total amount of water supplied to the crops from first watering to last watering is stored on land without any loss, then there will be a thick layer of water standing on the land. This depth of water is known as delta for the crop. It is denoted by Δ and expressed in cm.

Base period:

Base period is defined as the period from the first to last watering of the crop just before its maturity. It is denoted by B and expressed in number of days.

In open channel flow, the relationship between **duty (D)** and **delta (Δ)** is given by:

$$\Delta = \frac{8.64 \times B}{D}$$

or

$$D = \frac{8.64 \times B}{\Delta}$$

Where:

Δ (**Delta**) = Depth of water required by the crop during its base period (m)

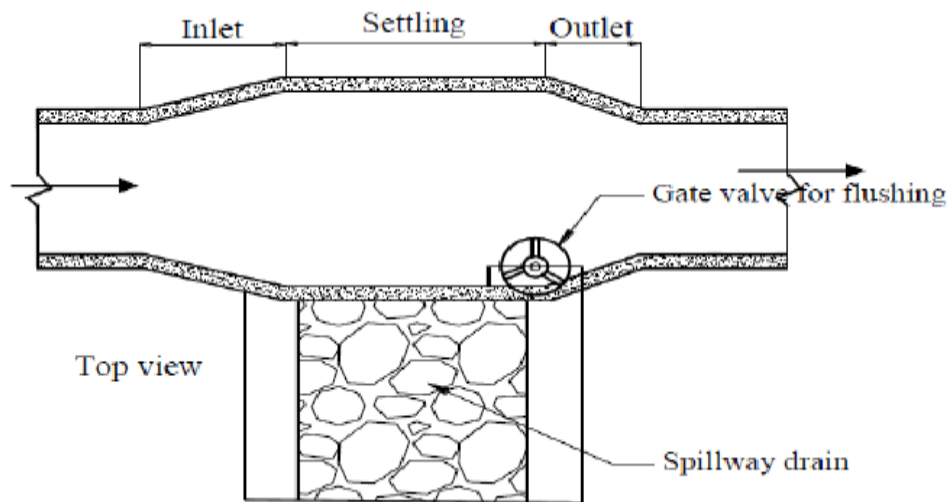
D = Duty of water (hectares/cumecs)

B = Base period of the crop (days)

7. In hydropower structures, what is the purpose of de-sanding basin? Explain its types and factors to be considered in its design. [3+3+4=10]

Answer:

De-sanding basin also known as de-stilling basin is a hydraulic structure placed between intake and penstock in the hydropower project for removal of suspended sediment particles from flowing water which cannot be trapped by the gravel trap.



Purposes of de-sanding basin:

i. Protection of turbine:

De-sanding basin ensures that the water entering the conveyance system is free of sediment particles so that, wear and damage on the turbine blades and other part of the turbines are prevented.

ii. Increasing the efficiency of hydropower project:

De- sanding basin increases the efficiency of the hydropower project by minimizing the damage on the structural component of the hydropower project.

iii. Consistent power generation:

Descending basin also encourages the consistent power generation as it reduces maintenance requirements and need for frequent replacement of damaged parts of the project.

Types of de-sanding basin:

i. On the basis of the direction of flow:

a. Horizontal flow de-sanding basin:

- Water flows horizontally at low velocity due to which the sediments settle at the bottom due to gravity.
- It is the most commonly used type of de-sanding basin, simple in construction and operation.
- Suitable for large discharges but requires larger area for its construction.

b. Vertical flow de-sanding basin:

- Water flows upward through the basin where heavier particles settle downward against the upward flow.

- It is used in the area where space for the basin is limited.
- However, it is difficult to maintain and operate.
- c. **Circular or vortex type de-sanding basin:**
 - Water enters tangentially and creates circular motion where centrifugal force helps to remove sand particles.
 - The design of de-sanding basin is complex but is efficient in removal of sediments.
 - Construction cost is higher.
- ii. **On the basis of flushing operation:**
 - a. **Continuous flushing de-sanding basin:**
 - In this type of de-sanding basin, sediments are continuously removed through flushing channels.
 - It is commonly used in hydropower and irrigation projects carrying large sediment loads.
 - The basin is highly effective for high sediment concentration and removal without interruption in operation.
 - b. **Discontinuous flushing de-sanding basin:**
 - In this type of de-sanding basin, sediments are allowed to accumulate and are periodically removed by flushing.
 - It is the simple arrangement but requires temporary shutdown while cleaning.

The factors to be considered in the design of de-sanding basin are listed below:

- i. **Design discharge:**
 - De-sanding basin must be designed for the maximum expected flow rate.
 - The capacity of de-sanding basin should be adequate to handle peak flows without reducing its efficiency.
- ii. **Size of particles:**
 - The size of particles required to be removed must be selected.
 - Generally, particles with its size greater than 0.15-0.2 mm are targeted for removal.
- iii. **Settling velocity of particles:**
 - It can be determined using stoke's law.

$$V_s = \frac{g(\rho_s - \rho)d^2}{18\mu}$$

Where:

V_s = settling velocity

d = particle diameter

ρ_s = density of particle

ρ = density of water

μ = dynamic viscosity

- iv. **Basin dimensions:**
 - The length should be adequate for complete settling of particles.
 - Length-to-width ratio is generally kept between **4:1 and 8:1**.

Section-C (25 Marks)

8. Illustrate hair pin bend in hill roads with neat sketch. [5]

Answer:

The road passing through the hilly terrain with a cross slope of 25% or more is generally termed as Hill Road. A hill road usually consists of either a river route or a ridge route. The steep, unstable slopes and complex geology of the Himalayas and Middle Hills make the design and construction of hill roads more critical. Hill roads are also provided with hair pin bends and sharp curves wherever required.

Hairpin bends are those provided in hill road alignment introduced when it becomes necessary to attain height without covering substantial horizontal distance. They may be designed as either a circular curve with transition at each end or as a compound circular curve. In hairpin bends,

Minimum design speed= 20 kmph

Minimum radius of curvature = 15 m

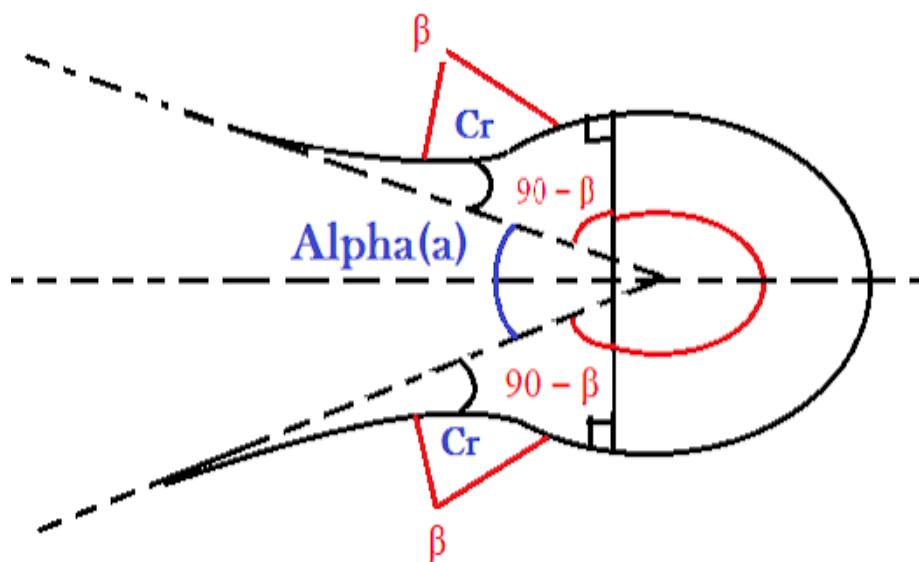
Minimum length of transition curve = 15 m

Maximum longitudinal gradient= 4%

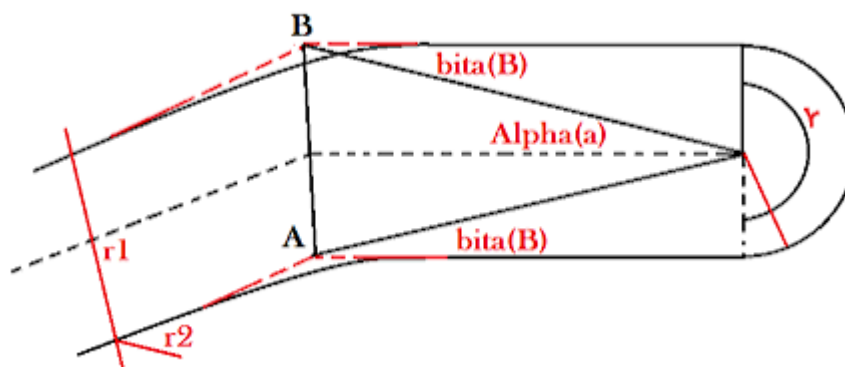
Maximum superelevation =10%

At the hairpin bends, inner and outer edges of the roadway should be concentric with respect to the centerline of the pavement. A minimum intervening distance of 60 m should be provided between successive bends to enable driver to negotiate the alignment smoothly, where a number of hair pin bends have to be introduced. At hairpin bends preferably the full roadway width should be surfaced.

However, hairpin bends are not the desirable elements of hill road as the speed have to be substantially reduced. The construction cost also substantially increases because of the extensive volume of earthworks and provision of retaining structures.



Type – I



Type- II

9. **What is highway drainage? Briefly describe the subsurface drainage for lowering of water table, control of seepage flow and control of capillary rise with typical sketches. [1+3+3+3]**

Answer:

Highway drainage can be defined as the process of interception and removal of water from over, under and vicinity of road surface. It can be surface drainage or subsurface drainage. Removal and diversion of surface

water from highway and adjoining land is called surface drainage. Similarly, removal of water from below the ground surface is called subsurface drainage.

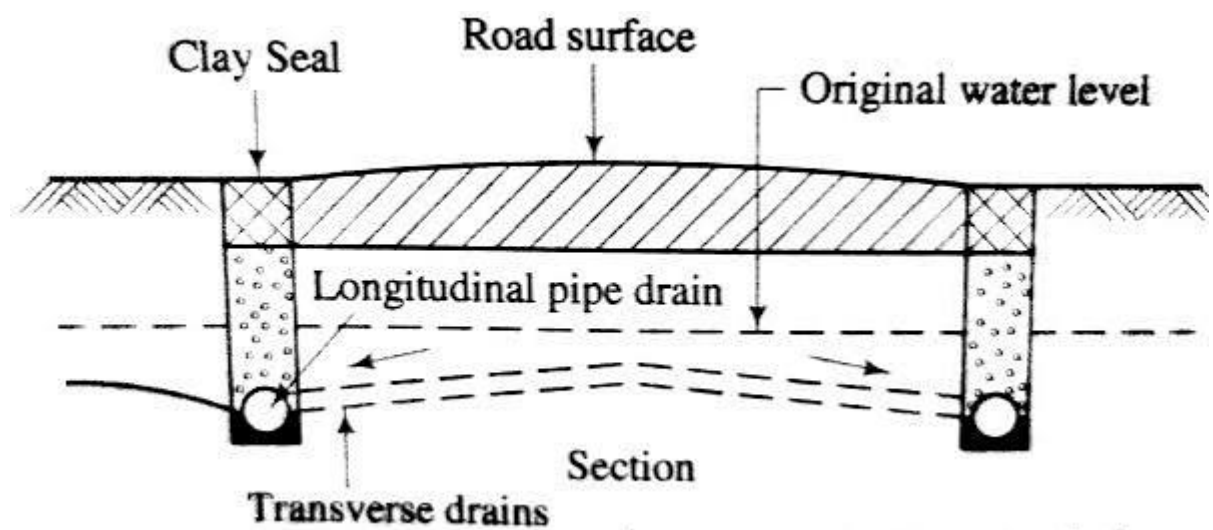
Variation in moisture content of subgrade is caused by seepage of water, penetration of moisture through the pavement, rise and fall of ground water table and many more. The subsurface drainage for lowering of water table, control of seepage flow and control of capillary rise are described below:

i. Lowering of GWT:

If the underground water table is more than 1.2 m below the subgrade of the road, it does not require any subsoil drainage. But if it is closer, the road formation is raised so that the height of subgrade remains 1.2 m above the highest water table.

If the soil is relatively permeable, then water table can be lowered by construction of longitudinal drainage trenches with drain pipes and filter sand.

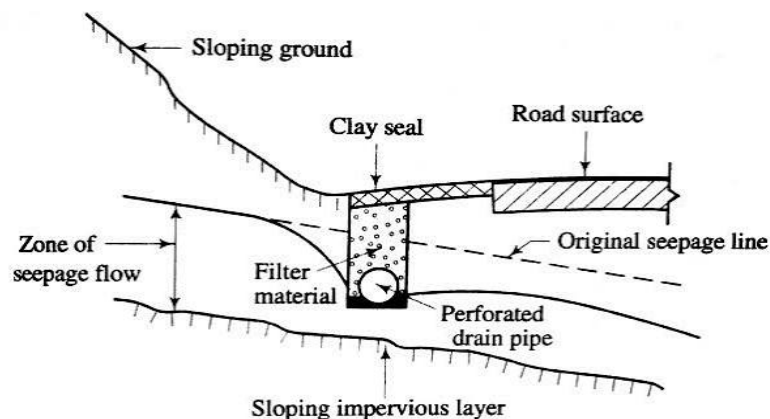
However, if the soil is relatively impermeable, then additional transverse drains are also provided for effective drain off of water.



ii. Control of seepage:

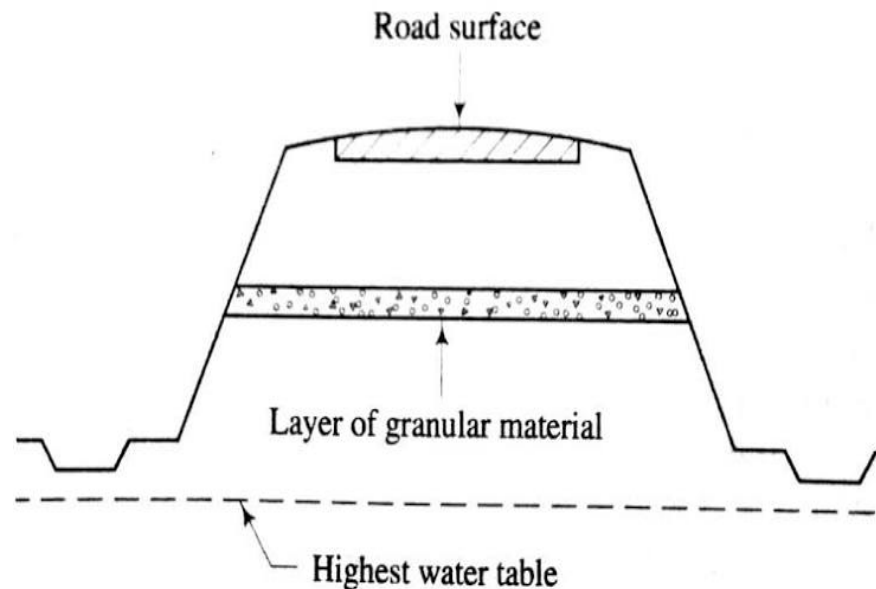
Seepage flow may reach the soil subgrade and affect its strength characteristics if the ground surface and the embedded impervious layer are sloping towards the road.

If the seepage level reaches the depth of less than 60 cm to 90 cm from the road subgrade it must be intercepted to keep seepage line below road subgrade.



iii. Control of capillary rise:

Capillary water also affects the strength of subgrade which can be prevented by lowering GWT if the subgrade soil is permeable type. However, for the impermeable subgrade soil, control of capillary rise can be done by capillary cutoffs.



10. Explain in brief the various impacts of airport in the vicinity area regarding aircraft noise, air quality and water and soil pollution. How these things are considered in airport planning. [6+4=10]

Answer:

Airport provides the safest and fastest mode of transportation, with significant economic benefits. However, construction of airport also has negative impacts on the environment and the area nearby the airport. The major impacts of airport can be aircraft noise, air quality, water pollution and soil pollution.

i. Aircraft noise impact:

- Aircraft generates large noises during takeoff, landing and taxiing which causes annoyance, stress, sleep disturbance in the nearby residents.
- It affects noise-sensitive areas such as schools, hospitals etc. and may give rise to mental and physical health problems to the people residing nearby

Consideration of airport noise impact on the airport planning:

- Airport sites must be selected away from densely populated areas.
- Runways should be properly oriented so that minimum flights fly over residential areas.
- Green belts plantation as noise barriers.
- Noise contour mapping and land use zoning.

ii. Air quality impact:

- Aircraft emits harmful gases and vapors such as carbon monoxide, nitrogen oxide, hydrocarbons from their engines and other ground support equipment.
- These pollutant causes air pollution and also contribute to greenhouse gases emissions.

Considerations of air quality impact on the airport planning:

- EIA must be conducted before construction.
- Promotion of energy efficient equipment and cleaner fuels for aircraft engines and ground support equipment.
- Airport facilities must be constructed separated from residential areas.

iii. Water pollution impact:

- Airport also contributes to water pollution as the aircraft may spill fuels and soils.
- It can also occur due to stormwater runoff carrying pollutants and harmful chemicals during aircraft washing operations or carrying out different maintenance activities.

Considerations of water pollution impact on the airport planning:

- Proper surface and subsurface drainage must be provided at the airport and adjacent areas.
- Proper measures must be taken for emergency oil spills.
- Regular monitoring of drainage systems and groundwater quality must be carried out.

iv. Soil pollution impacts:

- Airport also contribute to soil pollution due to leakage of fuels, lubricants and chemicals mixed to the soil.
- It may cause contamination of surrounding land and vegetation and might also affect soil fertility.

Consideration of soil pollution impacts in airport planning:

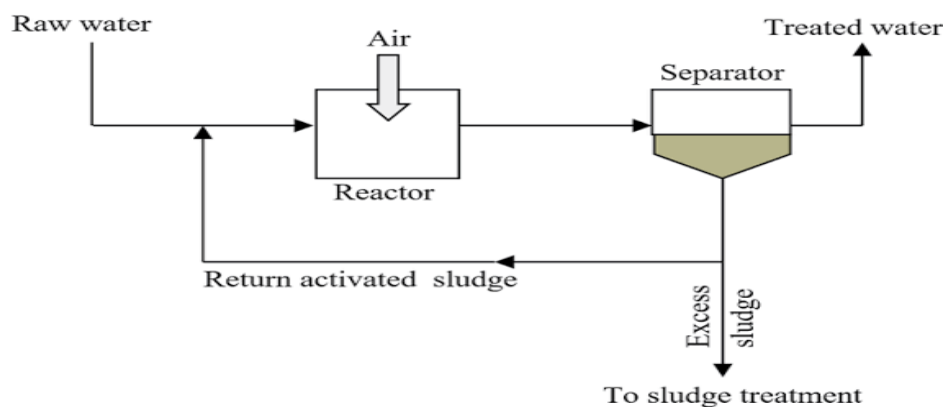
- Proper management of hazardous waste.
- Designated furl storage and handling areas must be ensured.
- Regular environmental monitoring and remedy measures must be carried out.

Section-D (25 Marks)

11. What is activated sludge processes in sewage treatment? Why BOD removal/ treatment for domestic waste is important before discharging in natural rivers. [5+5=10]

Answer:

Activated sludge process is a biological method of wastewater treatment that utilizes a mixture of microorganisms such as bacteria, fungi, protozoa and rotifers maintained in suspension, in an aerobic environment to metabolize organic matter to produce secondary wastewater effluent of acceptable quality. It has removal efficiencies in the range of 95-98% than trickling filters of 80-85%.



The main stages involved in activated sludge process is described below:

i. Physiochemical adsorption:

The fine suspended particles of the sewage combine with bacteria to form a sub layer of bacterial film at the surface. The film attracts finely divided colloidal and dissolved organic matter and result in formation of flocs. The flocs get settled at the bottom of sedimentation tanks and are removed.

ii. Oxidation:

Large flocs formed act as a vehicle for aerobic bacteria that oxidizes the carbonaceous and organic matter of wastewater by biochemical action. Settled microorganisms are removed from the bottom of settling tank are and are returned to aeration tank to metabolize additional organic matter. Balance between food and mass of the organisms, also known as sludge loading ratio must be maintained.

Working:

- A primary clarifier may be introduced to remove part of suspended solid present in the influent and this reduces the organic load to the activated sludge system.

- The biological aerator is filled with a mixer of activated sludge and influent known as mixed liquor. It is necessary to maintain certain liquid suspended solid (MLSS) in the aerated tank which maintain good removal efficiency.
- The aerator equipment transfers oxygen into the reactor for the oxidation of organic matter.
- The continuous introduction of new influent results in a continuous discharge of mixed liquor to the secondary settling tank where separation of solid and liquid takes place.
- The liquid leaves the system as a treated effluent whereas some part of this sludge is recirculated to aeration tank called as return sludge and rest is taken for anaerobic digestion.

Biochemical Oxygen Demand (BOD) is the amount of dissolved oxygen required by microorganisms to decompose biodegradable organic matter present in wastewater.

BOD treatment is necessary for the domestic waste before discharging in natural rivers because:

i. Prevents the depletion of natural resources:

Domestic sewage consists of large amount of organic matter. The discharge of untreated domestic sewage directly into river pollutes the natural resources. Microorganisms consumes the dissolved oxygen to decompose organic matter present in the sewage. This increases the BOD of sewage which rapidly decrease the DO content of river. Hence, to prevent depletion of natural resources, BOD treatment is necessary.

ii. Protects aquatic life:

Due to low DO level, aquatic life can be threatened as they require sufficient DO for their survival. Severe oxygen depletion results in destruction of aquatic ecosystems. Hence, domestic sewage should be adequately treated before disposing it to rivers.

iii. Maintains ecological balance:

High BOD leads to growth of micro-organisms and bacteria which disturbs the natural balance of ecosystem. Hence, disposal of domestic sewage only after treatment must be seriously considered.

iv. Prevents septic condition:

Excessive organic matter in sewage creates anaerobic conditions which gives rise to foul smell due to production of hydrogensulphide, methane and ammonia. This makes river water unpleasant and unsuitable for utilization. Hence, domestic sewage must not be directly disposed into river.

v. Prevents the spread of epidemics and diseases:

Polluted rivers may lead to spread of diseases and epidemics. When the polluted water from the river is used in any way by human or cattle, the pathogens and harmful contaminants present in the water might put serious effect on the health.

The necessity of BOD treatment can be observed in the Bagmati and Bishnumati rivers of Kathmandu Valley. The direct discharge of untreated domestic and industrial wastewater into these rivers has severely deteriorated their water quality. While the rivers are relatively clean upstream, they become turbid, polluted, and foul-smelling as they pass through Kathmandu. This highlights the importance of treating wastewater and reducing BOD before its discharge into natural water bodies. Delay in action might cost the loss of another natural resource and disbalance in the ecosystem.

12. Explain the need of EIA, IEE and BES in road and bridge projects. What is the provision of preparation and approval of EIA in government rules and regulations? [4+6=10]

Answer:

As per EPA 2076 and EPR 2077, there are three types of environmental studies required before any construction activities. They are:

i. Basic Environmental Study (BES):

- BES is conducted to identify the possible environmental impacts of a small-scale project and to recommend measures for preventing those impacts.
- Need of BES in road and bridge projects:
 - a. BES recommends simple mitigation measures to reduce the adverse effects.
 - b. It is done for projects that are likely to have only minor environmental effects.
 - c. It ensures compliance with environmental regulations at minimum cost and time.
- BES is done for
 1. length of road less than 5 km and construction of local roads.
 2. up to 100 m length of bridges.
- ii. Initial Environment examination (IEE):**
 - IEE is conducted to determine whether a proposed medium scale project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts.
 - If IEE Report submitted is approved by concerned authority, the project is implemented.
 - Need of IEE in road and bridge projects:
 - a. IEE determines whether the impacts can be mitigated through practical measures.
 - b. it is required for medium scale projects that has moderate effect on environment.
 - c. It also helps in assisting decision-makers in selecting environmentally acceptable alternatives.
 - d. It evaluates the impacts on land, water resources, forests, wildlife and socio-economic conditions.
 - IEE is done for:
 - a. Highways with the length of road 5-50 km
 - b. Construction of bridges with its length greater than 100m.
- iii. Environmental Impact assessment (EIA):**
 - It is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.
 - Need of EIA in road and bridge projects:
 - a. EIA provides detailed assessment of physical, biological, social, cultural and economic impacts.
 - b. EIA is required for large-scale development projects with significant environmental impacts that cannot be adequately assessed through a Brief Environmental Study (BES) or Initial Environmental Examination (IEE).
 - c. EIA also examines project alternatives, including design and selection of route alignment.
 - d. It also develops comprehensive environmental management and monitoring plans.
 - EIA is done for:
 - a. Highways with the length of road greater than 50 km
 - b. Road tunnels with the length greater than 3 km.

Provision of preparation and approval of EIA in government rules and regulations:

i. Screening:

- This step determines whether the proposed project requires IEE or EIA.
- No project requiring EIA can be implemented without approval of the environmental study report.

ii. Preparation of Scoping Document:

The proponent prepares Scoping Document following the format in **Schedule-5** and is submitted to the concerned approving authority.

- For federal projects, it is submitted to the **Concerned Ministry** (e.g., the Ministry of Irrigation or Physical Infrastructure).
- The Concerned Ministry reviews it, adds its opinion, and forwards it to the **Ministry of Forests and Environment (MoFE)** within **15 days**.
- MoFE must approve or modify the Scoping Report within **15 days** of receiving the recommendation.

iii. Preparation and Approval of work schedule:

- The proponent prepares a **Work Schedule** (per **Schedule-8**) that outlines the methodology and timeline for the study.
- The Scoping Report and Work Schedule are often submitted and approved together.
- Once reviewed, MoFE must approve the Work Schedule within **15 days**.

iv. Field study and Public Hearing:

- The study must be carried out by a team of qualified experts who meet the academic and experience criteria listed in **Schedule-13**.
- The proponent **must organize at least one public hearing** in the project area to gather feedback from local communities, forest user groups, and local representatives.
- The proponent **must publicize the hearing** via local radio, newspapers, or other media, and post notices at the ward office. Documentation of the hearing (attendance, suggestions, photos, and audio-visual recordings) must be included in the final report.

v. EIA Report Preparation:

- EIA report is prepared based on the approved Scoping Report, Work Schedule, and the suggestions from the public hearing according to the detailed format in **Schedule-12**.
- **Public Notice for Feedback:** The proponent must issue another **7-day notice** (Schedule-9) in a national daily newspaper to gather written suggestions on the potential environmental impacts of the project.
- Reports are generally prepared in **Nepali** language.
- For foreign-invested projects, they may be in English, provided a concise **executive summary in Nepali** is included.

vi. Recommendation and Submission:

- The proponent must include a formal **Recommendation Letter** (**सिफारिस पत्र**) from the concerned **Local Level** and the relevant thematic office (per **Schedule-14**). These bodies must provide the recommendation within **15 days**.
- **Submission:** For federal projects, the report is submitted to the **Concerned Ministry**.
- The Concerned Ministry reviews the report and forwards it to **MoFE** with its formal opinion within **15 days**.
- The final report must be submitted within **two years** of the Scoping/Work Schedule approval.

vii. Final Review and Approval of EIA:

- **Final Public Review:** Upon receiving the EIA report, MoFE must make it public for **7 days** on its website and via a national daily newspaper to collect final public comments.
- **Review Committee:** MoFE may form a **committee** of representatives and experts to investigate the report, which may include on-site inspections or monitoring.
- **Approval Decision:** If the investigation shows the project will not cause significant adverse environmental impacts, MoFE shall approve the EIA report within **35 days**. The approval may include specific, binding conditions that the proponent must follow.
- Additionally, the project must start within **three years** of approval, or the proponent may be required to conduct a new study.

Note on Violations: Implementing a project without an approved EIA report or in violation of its conditions can lead to a fine of up to **5 million (५० लाख) rupees**.